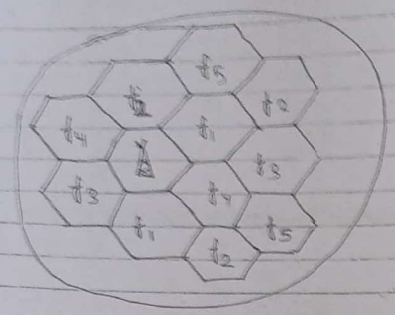


Fig: Cellular network.

→ In cellular system, each cell has a base transceiver (transmitter + receiver). The transmission power is controlled to allow commⁿ within the cell using a given freqⁿ while limiting the power at that freqⁿ that escapes the cell into adjacent ones.

→ Cellular system provide concept of freqⁿ reuse.

→ Support multiple users.

Chapter 7

Impairments, Error handling & Compression Technique

1) Impairments

→ In communication system, the signal that is received may differ from the signal that is transmitted due to various transmission impairments.

→ For analog signals, these impairments can degrade the signal quality.

→ For digital signals, bit errors (i.e altering of bits) may be introduced.

1) Attenuation

→ Attenuation means a loss of energy. When a signal travels through a medium, its strength decreases with distance.

→ Attenuation is generally exponential & thus is typically expressed as a constant number of decibels per unit distance.

→ Amplifiers (to amplify signal) or repeaters (to regenerate signal) are use to increase the signal strength.

→ Beyond a certain distance, the attenuation becomes unacceptably great, & repeaters or amplifiers are used to boost the signal at regular intervals.

2) Distortion

→ Distortion means that the signal changes its form or shape. Distortion can occur in a composite signal made of different frequencies.

→ Each signal component has its own propagation speed through a medium & therefore, its own delay in arriving at the final destination.

→ Differences in delay may create a difference in phase

if the delay is not exactly the same as the period duration.

→ Signal components at the receiver have phases different from what they had at the sender. The shape of the composite signal is therefore not the same.

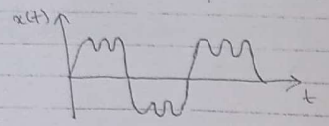


Fig: Composite signal at sender

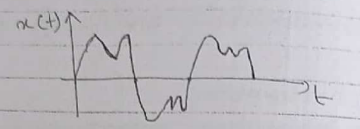


Fig: Distorted composite signal at receiver

• Delay Distortion

→ Delay Distortion occurs because the velocity of propagation of a signal through a guided medium varies with freq^n.

→ For bandlimited signal the velocity tends to be highest near the center freq^n & fall off towards the two edges of the band. Thus various freq^n components of a signal will arrive at the receiver at different times, resulting in phase shifts bet^n the different frequencies.

→ Delay distortion is critical for digital data. Because of delay distortion, some of the signal components of one bit position will spill over into other bit positions (overlapping), causing intersymbol interference which limits bit rate over a transmission channel.

• Noise

Noise may be defined as an unwanted form of energy which tend to interfere with the proper reception & reproduction of transmitted signals.

Noise

- External noise
  → noise which is generated external to a communication system.
  e.g atmospheric noise, Industrial noise, galatic noise.
- Internal noise
  → noise which is generated internally or within the comm^n system.
  e.g Shot noise, Thermal noise.

Galatic noise → due to cosmic ray from galaxies.

Shot noise → arises in active devices due to random behaviour of charge carriers. In electron tubes, shot noise is generated due to the random emission of electrons from cathodes, whereas in semiconductor devices shot noise generated due to the random diffusion of minority carriers or simply random generation & recombination of electron-hole pairs.

Thermal noise → also called white noise or Johnson noise.
→ is the random noise which is generated in a resistor or the resistive component of a complex impedance due to rapid & random motion of the molecules, atoms & electrons.

$P_n = K \cdot T \cdot B$ Expression for noise power.

K → Boltzmann's const $= 1.38 \times 10^{-23}$ Joule/deg·K
T → absolute temp.

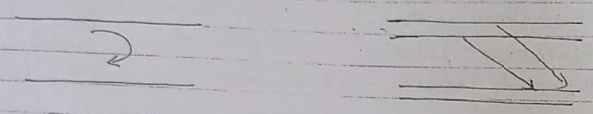
• Crosstalk

→ It is an unwanted coupling between signal paths. It can occur by electrical coupling between nearby twisted pairs or coaxial cable (rarely) lines carrying multiple signals.

→ In case of real life experience of crosstalk is while using the telephone, we hear another conversation.

Two types

1) Near End Crosstalk (NEXT)
2) Far End Crosstalk (FEXT)



NEXT — inside single channel.

FEXT — betn two transmission channel

• Interference

It is the type of signal impairment in which one signal interfere the freqn component of another signal. Here the freqn component of two signals interfere eachother & loss of certain components.

Types of Errors

In digital transmission systems, an error occurs when a bit is altered between transmission & reception i.e a binary 1 is transmitted & a binary 0 is received, or, a binary 0 is transmitted & a binary 1 is received.

Two general types of errors can occur: single-bit errors & burst errors.

→ A single-bit error is an isolated error condition that alters one bit but does not affect nearby bits.

→ A burst error of length B is a contiguous sequence of B bits in which the first & last bits & any number of intermediate bits are received in error.

Error detection techniques

Parity Coding

The simplest method of coding is a parity coding. In this method, parity bits are added to the message at the time of transmission. At the reception end, receiver checks these parity bits. Errors are detected if the expected pattern of parity bits is not received.

Parity may be of two types:

i) Even Parity

The parity of a binary word is known as 'even' if it contains even number of 1's. e.g 01101001 has even parity since there are four number of 1's in the word.

ii) Odd Parity

The parity of a binary word is known as 'odd' if it contains odd number of 1's. e.g 01001001.

Vertical Redundancy Check (VRC)

In this method, the parity check is done on each character seperately. In ASCII code, each character has seven binary digits. Also, one redundant digit is added to each character to make the parity odd or even.

Character	ASCII code of the character								Additional parity bit
	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8	(even parity)
K	1	1	0	1	0	0	1	0	0
O	1	1	1	1	0	0	1	1	0
I	1	0	0	1	0	1	1	1	1
⋮									

→ During the transmission, if single error is created, it will make number of 1's odd.

e.g

	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8
Transmitted message for character K (Even parity)	1	1	0	1	0	0	1	0
Received message	1	1	0	1	1	0	1	0

error digit.

→ If two digits are in error, then number of 1's in the received message is even. In this case, receiver will detect no error. This is the main drawback of VRC.

→ VRC check error only in one direction i.e either in row or in column.

Longitudinal Redundancy Check (LRC)

→ In LRC, large message block is divided into several characters & parity check is applied in both row & columns. Thus, it is possible to detect upto tripplen errors.

→ It is also possible to correct single errors with LRC.

→ In LRC for checking the message block, a complete character known as 'Block check character' BCC is added at the end of block of information. BCC can be even or odd parity

→ The structure of information block is as shown below.

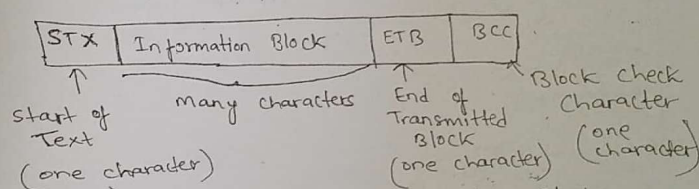


Fig: Structure of Information Block.

Characters	STX	N	A	E	T	E	S	ETB	BCC ← Column Parity bits
Msg Bits b_1	0	0	1	1	0	1	1	1	1
b_2	1	1	0	0	0	0	1	1	0
b_3	0	1	0	0	1	0	0	1	1
b_4	0	1	0	1	0	1	0	0	1
b_5	0	0	0	0	1	0	1	1	1
b_6	0	0	0	0	0	0	0	0	0
b_7	0	1	1	1	1	1	1	0	0
Row Parity LRC → b_8	1								

	N	A	I	-	I	S			
b_1	0	0	1	1	0	1	1	1	
b_2	1	1	0	0	0	0	1	0	
b_3	0	1	0	0	1	0	0	0	
b_4	0	1	0	1	1	1	0	1	X
b_5	0	0	0	0	1	0	1	1	
b_6	0	0	0	0	0	0	0	0	
b_7	0	1	1	1	1	1	1	0	
b_8	1	0	0	1	1	0	1	1	
Row Parity Bits					X				

Fig: Detected message Block.
X implies an error in the parity.

Advantages of LRC over VRC
→ can correct single error.
→ can detect double or tripple errors.

Draw back
→ complex.

Cyclic Redundancy check
→ Use to correct error.
→ CRC codes with C(7,4) has 4 dataword & 7 Codeword.
→ In the encoder, the dataword has k bits (4 here) & the codeword has n bits (7 here). The size of the dataword is augmented by adding n-k (3) zeros (0s) to the right side of the word.
→ The generator divides the augmented dataword by the divisor (modulo-2 division). The quotient of the division is discarded & the remainder ($r_2 r_1 r_0$) appended to the dataword to create the codeword.

→ The decoder receives the possibly corrupted codeword. A copy of all n bits is fed to the checker which is a replica of the generator. The remainder produced by the checker is a syndrome of n-k (3 here) bits, which is fed to the decision logic analyzer.
→ If the syndrome bits are all 0s, the 4 left most bits of the codeword are accepted as the dataword (interpreted as no error); otherwise, the 4 bits are discarded (error).

e.g. Dataword → 1001
Divisor → 1011

```
        1001
1011 | 1001 000 |
       1011
        0100
        0000
         1000
         1011
          0110
          0000
           110  ← Remainder
```

codeword $\boxed{1001}\boxed{110}$

Codeword ⇒ 1001 110

```
        1010
1011 | 1001 110 |
       1011
        0101
        0000
         1011
         1011
          0000
          0000
           000
```

Syndrome

Dataword accepted

If codeword = 1000110

```
1011 | 1000110
       1011
        0111
        0000
         1111
         1011
          1000
          1011
          011   Syndrome
```

Data word discarded.

Checksum

→ Like CRC, checksum is also based on the concept of redundancy.

Sender Site:

1) The message is divided into 16-bit words.
2) The value of the checksum word is set to 0.
3) All words including the checksum are added using 1's complement addition.
4) The sum is complemented & becomes the checksum.
5) The checksum is sent with the data.

Receiver Site:

1) The message including checksum is divided into 16 bit words.
2) All words are added using one's complement addition.
3) The sum is complemented & becomes the new checksum.
4) If the value of checksum is 0, the message is accepted; otherwise it is rejected.

e.g 10101001 00111001

Transmitter

Data	Segment 1	10101001
	Segment 2	00111001
		00000000
	Sum	11100010
	1's complement	00011101 ← checksum

Receiver

```
10101001
00111001   } → Data
11111111 ← sum
00000000 ← 1's Complement.
```

Data accepted.

* If MSB of Received data segment 1 is inverted then,

```
00011101 ← Checksum
00101001 } → Data
00111001
01111111 ← sum
10000000 ← 1's complement.
```

Data rejected.
error in 1st bit.
error detected.

To calculate r_1 → compute even parities of the association

		Parity	
3, 5, 7, 9, 11			
1 1 0 0 1		1	r_1
r_2 → 3, 6, 7, 10, 11			
1 1 0 1 1		0	r_2
r_4 → 5, 6, 7			
1 1 0		0	r_4
r_8 → 9, 10, 11			
0 1 1		0	r_8

Receiver

11	10	9	8	7	6	5	4	3	2	1
1	1	0	0	0	1	1	0	1	0	1

		Parity bit
r_1 → 1, 3, 5, 7, 9, 11	1 1 1 0 0 1	0
r_2 → 2, 3, 6, 7, 10, 11	0 1 1 0 1 1	0
r_4 → 4, 5, 6, 7	0 1 1 0	0
r_8 → 8, 9, 10, 11	0 0 1 1	0

No error.

If 7th bit inverted during transmission.

11	10	9	8	7	6	5	4	3	2	1
1	1	0	0	1	1	1	0	1	0	1

		Parity bit
r_1 → 1, 3, 5, 7, 9, 11	1 1 1 1 0 1	1
r_2 → 2, 3, 6, 7, 10, 11	0 1 1 1 1 1	1
r_4 → 4, 5, 6, 7	0 1 1 1	1
r_8 → 8, 9, 10, 11	0 0 1 1	0

$r_8\, r_4\, r_2\, r_1$ → 0111 i.e error in 7th bit.

Types of Data Compression techniques

→ To send audio or video over the Internet, we require compression techniques. There are different types of audio compression & video compression.

→ It is usually acceptable to have the audio or video signal after encoding & then decoding be slightly different from original.

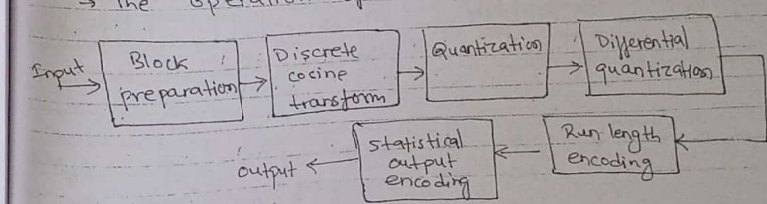
→ when the decoded output is not exactly equal to the original input, the system is said to be lossy & compression is lossy compression.

→ when the decoded output is exactly equal to the Original input, the system is said to be lossless & compression is lossless compression.

• JPEG Standard

→ Joint Photographic Experts Group is standard for compressing continuous-tone still pictures (i.e photographs) was developed by photographic experts working under ITU, ISO or other standards body.

→ The operation of JPEG is shown below



- MPEG Standard

→ Motion Picture Experts Group.

→ MPEG can compress both audio & video.

→ A motion picture is a rapid flow of a set of frames, where each frame is an image.

→ A frame is spatial combination of pixels & a video is a temporal combination of frames that are sent one after another.

Various MPEG standards

MPEG-1, MPEG-2, MPEG-3, MPEG-4

→ In all standard, there is spatial compression & Temporal compression.

Chapter 8

Data link Control & Protocol

The main responsibilities of the data link layer are framing, flow control, errors control and software implemented protocols that provide smooth & reliable transmission betⁿ nodes.

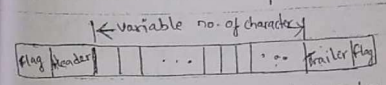
: Framing

Framing deals with how to organize the bits that are carried by the physical layer. frames can be of fixed or variable size.

Fixed size framing: There is no need for defining the boundries of the frames because the size itself can be used as a delimiter. e.g ATM cell.

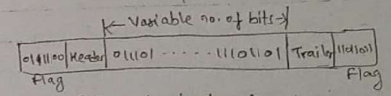
Variable size framing: The size of frames are variable. It may use variable number of character or variable number of bits depending on which two protocols are used.

Character-oriented protocol



Character oriented framing

Bit-oriented protocols



Bit oriented framing.

→ Byte stuffing is the process of adding 1 extra byte whenever there is a flag in the text.